

**SCHOOL OF COMPUTING**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**(SPECIALIZATION IN AI, BCT, CS & IoT)**  
**FINAL YEAR PROJECT-PHASE-1**  
**ZEROTH REVIEW**

**Course Code/ Name:**

**Batch ID:** \_\_\_\_\_

**Program:** BE-CSE    **Specialization:** CYBERSECURITY    **Batch:** 2023-2027

**Date:**

**Project Details**

|                        |                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------|
| <b>TITLE</b>           | QUBIT (Quantum Upgrade Bridge & Inventory Tool)                                                          |
| <b>PROJECT OUTCOME</b> | <input type="checkbox"/> Research Paper <input type="checkbox"/> Patent <input type="checkbox"/> Product |
| <b>GUIDE NAME</b>      | Dr.P.Shanmuga Prabha                                                                                     |

**Student Details**

| S.No | Name           | Register Number | No. of time came for Discussion | Approval for First Review |
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**Project Guide**  
Name & Signature with Date

**Internal Examiner**  
Name & Signature with Date

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# **QUBIT: A Quantum Upgrade Bridge & Inventory Tool for HNDL Risk Modeling and Automated Cryptographic Migration**

**Team Members Name with Register Number & Specialization:**

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**Area/Domain of Project:** Cyber Security, Post-Quantum Cryptography (PQC), Post-Quantum Migration

## **ABSTRACT**

The rapid advancement of quantum computing poses a significant threat to existing public-key cryptographic systems, enabling future adversaries to exploit the **Harvest Now, Decrypt Later (HNDL)** strategy by storing encrypted data today for decryption once cryptographically relevant quantum computers become available. This emerging risk threatens critical sectors such as government, healthcare, finance, and cloud infrastructure, making proactive migration to post-quantum cryptography essential. **QUBIT (Quantum Upgrade Bridge & Inventory Tool)** is proposed to address this challenge by providing an intelligent, automated, and open-source framework for identifying vulnerable cryptographic assets and facilitating secure migration to quantum-resistant algorithms. Existing solutions primarily offer migration guidelines or expensive commercial products but lack integrated risk assessment, automated cryptographic inventory, migration prioritization, and hybrid transition support. **QUBIT** bridges these gaps by combining cryptographic asset discovery, Bayesian risk quantification, Monte Carlo-based quantum timeline estimation, AI-assisted migration planning, and a hybrid classical–post-quantum cryptographic bridge within a unified platform. The proposed framework enables organizations to prioritize vulnerable assets, generate migration recommendations, and ensure a smooth transition while maintaining operational continuity. By improving cryptographic visibility, reducing migration complexity, and strengthening long-term cybersecurity resilience, **QUBIT** supports organizations in



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preparing for the post-quantum era through a scalable, cost-effective, and research-driven solution.

**Keywords:** Post-Quantum Cryptography, Harvest Now Decrypt Later (HNDL), Cryptographic Asset Discovery, Quantum Risk Assessment, Automated Cryptographic Migration

**Signature of Student**

**Signature of Guide**